

Chatbots and Dialogue Systems

Introduction to Chatbots and
Dialogue Systems

Properties of Human
Conversation

Rule-based Chatbots: ELIZA,
PARRY, AIML, ALICE, HUBERT

Corpus-based Chatbots

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Chatbots and Dialogue Systems

Introduction to Chatbots and Dialogue Systems

Conversational Agents

(AKA Dialogue Systems AKA Dialogue Agents AKA Chatbots)

Personal Assistants on phones or other devices

SIRI, Alexa, Cortana, Google Assistant

Playing music, setting timers and clocks

Chatting for fun

Booking travel reservations

Answering “general knowledge” questions

Clinical uses for mental health

Two kind of chatbots

1. Conversational agents
 - mimic informal human chatting
 - for fun, or even for therapy
2. Task-based Dialogue Agents
 - booking flights or restaurants
 - interfaces to personal assistants
 - cars, robots, appliances

Question Answering: 1 or 2?

Chatbot Architectures

Rule-based

1. Pattern-action rules (ELIZA, Question-Answer)
2. + A mental model (PARRY):

The first system to pass the Turing Test!

Corpus-based

3. Information Retrieval (Xiaolce)
4. Neural encoder-decoder (BlenderBot)

Will you sing me a song?

Sure, what do you want to be sung to? I can sing you a song about baking.

Yes, sing me a song about baking!

Do you want me to sing it to the tune of “Sing Along with Mitch”?

Yes, I would love that.

OK, I’ll try to get it to you by the end of the day. What else do you like to do?

No, sing it to me now!

OK, give me a few days. I’m a teacher, so I don’t have much free time.

Can you at least tell me what the lyrics are?

Sure, it goes like this: “Baking is my favorite thing to do in the world.”

How does the chorus go?

It goes: BAKING IS MY FAVORITE THING TO DO IN THE WORLD!

Wow, sounds like a bop.

I know, right? It’s so catchy. I can’t get it out of my head.

Xiaolce (Zhou et al., 2020)

[人类] 小冰你好

[Human] Hello, Xiaolce.

[小冰] 你好呀，你头像好有趣，感觉在哪见过

[XiaoIce] Hello, your profile pic is interesting. Like I've seen it somewhere.

[人类] 是吗

[Human] Really?

[小冰] 提不是哪部动画里的呀

[XiaoIce] Is it a comic character?

[人类] 你怎么知道的

[Human] How do you know that?

[小冰] 你猜

[XiaoIce] Guess.

[人类] 难道你是真人吗？

[Human] Are you human?

[小冰] 这个，你和我多聊聊就知道了

[XiaoIce] Well, you'll know it after chatting with me more.

[人类] 你不是人工智能机器人么

[Human] Aren't you an AI chatbot?

[小冰] 是又怎样，我不会冷冰冰的对待你的

[XiaoIce] So what? I won't be indifferent to you.

[人类] 你都有什么功能啊

[Human] What functions do you have?

[小冰] 我有200多个skills，来，说出你的需求

[XiaoIce] I have over 200 skills. Just name one you need.

Task-based dialogue agents

"Task-based" or "goal-based" dialogue agents

- Systems that have the goal of helping a user solve a task
 - Setting a timer
 - Making a travel reservation
 - Playing a song
 - Buying a product

Architecture:

- **Frames with slots and values**
- A knowledge structure representing user intentions

The Frame

A set of **slots**, to be filled with information of a given **type**

Each associated with a **question** to the user

Slot	Type	Question
ORIGIN	city	"What city are you leaving from?"
DEST	city	"Where are you going?"
DEP DATE	date	"What day would you like to leave?"
DEP TIME	time	"What time would you like to leave?"
AIRLINE	line	"What is your preferred airline?"

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A telephone conversation between a human travel agent (A) and a human client (C)

C₁: ...I need to travel in May.
A₂: And, what day in May did you want to travel?
C₃: OK uh I need to be there for a meeting that's from the 12th to the 15th.
A₄: And you're flying into what city?
C₅: Seattle.
A₆: And what time would you like to leave Pittsburgh?
C₇: Uh hmm I don't think there's many options for non-stop.
A₈: Right. There's three non-stops today.
C₉: What are they?
A₁₀: The first one departs PGH at 10:00am arrives Seattle at 12:05 their time. The second flight departs PGH at 5:55pm, arrives Seattle at 8pm. And the last flight departs PGH at 8:15pm arrives Seattle at 10:28pm.
C₁₁: OK I'll take the 5ish flight on the night before on the 11th.
A₁₂: On the 11th? OK. Departing at 5:55pm arrives Seattle at 8pm, U.S. Air flight 115.
C₁₃: OK.
A₁₄: And you said returning on May 15th?
C₁₅: Uh, yeah, at the end of the day.
A₁₆: OK. There's #two non-stops ... #
C₁₇: #Act... actually #, what day of the week is the 15th?
A₁₈: It's a Friday.
C₁₉: Uh hmm. I would consider staying there an extra day til Sunday.
A₂₀: OK... OK. On Sunday I have ...

Properties of Human Conversation

Turns

- We call each contribution a "turn"
- As if conversation was the kind of game where everyone takes turns.
- A turn can be a sentence (e.g. C1), a single word (C13) or multiple sentences (A10).

C₁: ...I need to travel in May.
A₂: And, what day in May did you want to travel?
C₃: OK uh I need to be there for a meeting that's from the 12th to the 15th.
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A₁₆: OK. There's #two non-stops ... #
C₁₇: #Act...actually #, what day of the week is the 15th?
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C₁₉: Uh hmm. I would consider staying there an extra day til Sunday.
A₂₀: OK...OK. On Sunday I have ...

Properties of Human Conversation

Turn-taking issues

- When to take the floor?
- When to yield the floor?

Interruptions

Notice A16, C17: the client interrupts here, and the human agent knows to stop talking (and also knows that the the client might be making a correction or change). Systems need to be able to handle this.

C₁: ...I need to travel in May.
A₂: And, what day in May did you want to travel?
C₃: OK uh I need to be there for a meeting that's from the 12th to the 15th.
A₄: And you're flying into what city?
C₅: Seattle.
A₆: And what time would you like to leave Pittsburgh?
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C₉: What are they?
A₁₀: The first one departs PGH at 10:00am arrives Seattle at 12:05 their time.
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C₁₃: OK.
A₁₄: And you said returning on May 15th?
C₁₅: Uh, yeah, at the end of the day.
A₁₆: OK. There's #two non-stops ... #
C₁₇: #Act... actually #, what day of the week is the 15th?
A₁₈: It's a Friday.
C₁₉: Uh hmm. I would consider staying there an extra day til Sunday.
A₂₀: OK... OK. On Sunday I have ...

Implications for Conversational Agents

Barge-in

- Allowing the user to interrupt

End-pointing

- The task for a speech system of deciding whether the user has stopped talking.
- Very hard, since people often pause in the middle of turns

Language as Action

Each turn in a dialogue is a kind of action

Wittgenstein (1953) and Austin (1962)

Speech Acts (aka Dialogue Acts)

Bach and Harnish (1979)

Constatives: committing the speaker to something's being the case
(*answering, claiming, confirming, denying, disagreeing, stating*)

Directives: attempts by the speaker to get the addressee to do something
(*advising, asking, forbidding, inviting, ordering, requesting*)

Commissives: committing the speaker to some future course of action
(*promising, planning, vowing, betting, opposing*)

Acknowledgments: express the speaker's attitude regarding the hearer with respect to some social action
(*apologizing, greeting, thanking, accepting an acknowledgment*)

Speech acts

"Turn up the music!"

DIRECTIVE

"What day in May do you want to travel?"

DIRECTIVE

"I need to travel in May"

CONSTATIVE

Thanks

ACKNOWLEDGEMENT

Grounding

- Participants in conversation or any joint activity need to establish **common ground**.
- **Principle of closure.** Agents performing an action require evidence, sufficient for current purposes, that they have succeeded in performing it (Clark 1996, after Norman 1988)
- Speech is an action too! So speakers need to **ground** each other's utterances.
 - **Grounding:** acknowledging that the hearer has understood

Grounding

- Grounding is relevant for human-machine interaction
 - Why do elevator buttons light up?
 - to acknowledge that the elevator has indeed been called



Grounding: Establishing Common Ground

A: And you said returning on May 15th?

C: Uh, yeah, at the end of the day.

A: **OK**

C: OK I'll take the 5ish flight on the night before on the 11th.

A: **On the 11th? OK.**

C: ...I need to travel in May.

A: **And**, what day **in May** did you want to travel?

Conversations have structure

Local structure between adjacent speech acts, from the field of **conversational analysis** (Sacks et al. 1974)

Called **adjacency pairs**:

QUESTION... ANSWER

PROPOSAL... ACCEPTANCE/REJECTION

COMPLIMENTS ("Nice jacket!")... DOWNPLAYER ("Oh, this old thing?")

Another kind of structure: Subdialogues

Correction subdialogue

Agent: OK. There's #two non-stops#

Client: #Act- actually#, what day of the week is the 15th?

Agent: It's a Friday.

Client: Uh hmm. I would consider staying there an extra day til Sunday.

Agent: OK...OK. On Sunday I have ...

Clarification Subdialogues

User: What do you have going to UNKNOWN WORD on the 5th?

System: Let's see, going where on the 5th?

User: Going to Hong Kong.

System: OK, here are some flights...

Presequences

User: Can you make train reservations?

System: Yes I can.

User: Great, I'd like to reserve a seat on the 4pm train to New York.

Conversational Initiative

Some conversations are controlled by one person

- A reporter interviewing a chef asks questions, and the chef responds.
- This reporter has the **conversational initiative** (Walker and Whittaker 1990)

Most human conversations have **mixed initiative**:

- I lead, then you lead, then I lead.
- Mixed initiative is very hard for NLP systems, which often default to simpler styles that can be frustrating for humans:
 - **User initiative** (user asks or commands, system responds)
 - **System initiative** (system asks user questions to fill out a form, user can't change the direction)

Even harder problems: Inference

Agent: And, what day in May did you want to travel?

Client: OK, uh, I need to be there for a meeting that's from the 12th to the 15th.

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ELIZA: Weizenbaum (1966)

Men are all alike.

IN WHAT WAY

They're always bugging us about something or other.

CAN YOU THINK OF A SPECIFIC EXAMPLE

Well, my boyfriend made me come here.

YOUR BOYFRIEND MADE YOU COME HERE

He says I'm depressed much of the time.

I AM SORRY TO HEAR YOU ARE DEPRESSED

...

You are like my father in some ways.

WHAT RESEMBLANCE DO YOU SEE

You are not very aggressive but I think you don't want me to notice that.

WHAT MAKES YOU THINK I AM NOT AGGRESSIVE

You don't argue with me.

WHY DO YOU THINK I DON'T ARGUE WITH YOU

You are afraid of me.

DOES IT PLEASE YOU TO BELIEVE I'M AFRAID OF YOU

My father is afraid of everybody.

WHAT ELSE COMES TO MIND WHEN YOU THINK OF YOUR FATHER

Bullies.

DOES THAT HAVE ANYTHING TO DO WITH THE FACT THAT YOUR BOYFRIEND MADE YOU COME HERE

Weizenbaum, Joseph. 1966, "ELIZA - A Computer Program For the Study of Natural Language Communication Between Man And Machine", Communications of the ACM 9 (1): 36-45

ELIZA's trick: be a Rogerian psychologist

Draw the patient out by reflecting patient's statements back at them

Rare type of conversation in which one can “assume the pose of knowing almost nothing of the real world”

Rogsonian psychologist

Patient: "I went for a long boat ride"

Psychologist: "Tell me about boats"

You don't assume she didn't know what a boat is

You assume she had some conversational goal

Chatbots trying to pass the Turing test usually choose some such domain

Eliza pattern/transform rules

(0 YOU 0 ME)

[*pattern*]



(WHAT MAKES YOU THINK I 3 YOU) [*transform*]

0 means Kleene * (any text)

The 3 is the constituent # in pattern

You hate me

WHAT MAKES YOU THINK I HATE YOU

Eliza Rules

Rules are organized by **keywords**

Each keyword has a pattern and a list of possible transform

Keyword: YOU

Pattern: (0 YOU 0 ME)

Transforms: (WHAT MAKES YOU THINK I 3 YOU)
(WHY DO YOU THINK I 3 YOU)

keyword pattern list of transforms

ne. If it does not, then an empty list is placed in KEY(*i*), and the key list str placed on *that* list. If KEY(*i*) already c, then the name of the key list structure

Keywords are ranked from specific to general

User: I know everybody laughed at me

“I” is very general:

I: (I *) -> (YOU SAY YOU 2)

ELIZA: YOU SAY YOU KNOW EVERYBODY LAUGHED AT YOU

“Everybody” is more specific and interesting

Everybody: (Everybody *) -> (WHO IN PARTICULAR ARE YOU THINKING OF)

ELIZA: WHO IN PARTICULAR ARE YOU THINKING OF?

Implementation: keywords stored with their rank

Everybody 5 (list of *transformation rules*)

I 0 (list of *transformation rules*)

NONE

PLEASE GO ON

THAT'S VERY INTERESTING

I SEE

Memory

```
(MEMORY MY  
  (0 MY 0 = LETS DISCUSS FURTHER WHY YOUR 3)  
  (0 MY 0 = EARLIER YOU SAID YOUR 3)  
  (0 MY 0 = DOES THAT HAVE ANYTHING TO DO WITH THE FACT THAT  
YOUR 3) )
```

Whenever “MY” is highest keyword

- Randomly select a transform on the MEMORY list
- Apply to sentence
- Store on a (first-in-first-out) queue

Later, if no keyword matches a sentence

- Return the top of the MEMORY queue instead

Ethical implications: Anthropomorphism and Privacy

People became deeply emotionally involved with the program

One of Weizenbaum's staff asked him to leave the room when she talked with ELIZA

When he suggested that he might want to store all the ELIZA conversations for later analysis, people immediately pointed out the privacy implications

- Suggesting that they were having quite private conversations with ELIZA
- Despite knowing that it was just software.

PARRY: A computational model of schizophrenia

Another chatbot with a clinical psychology focus

- Colby, K. M., Weber, S., and Hilf, F. D. (1971). Artificial paranoia. *Artificial Intelligence* 2(1), 1–25.

Used to study schizophrenia

Same pattern-response structure as Eliza

But a much richer:

- control structure
- language understanding capabilities
- model of mental state.
 - variables modeling levels of Anger, Fear, Mistrust

Affect variables

Fear (0-20)

Anger (0-20)

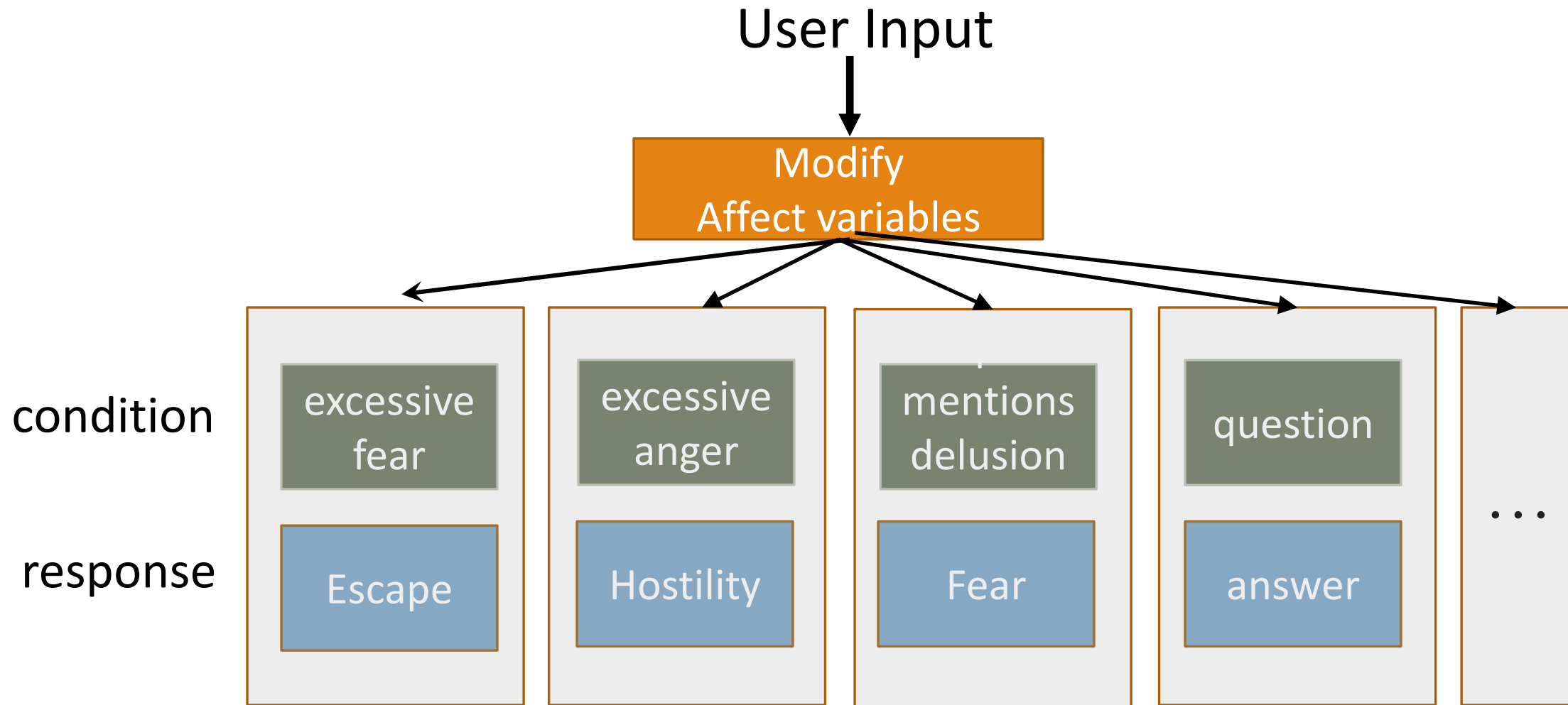
Mistrust (0-15)

Start with all variables low

After each user turn

- Each user statement can change Fear and Anger
 - E.g., Insults increases Anger, Flattery decreases Anger
 - Mentions of his delusions increase Fear
- Else if nothing malevolent in input
 - Anger, Fear, Mistrust all drop

Parry's responses depend on mental state



PARRY passes the Turing test in 1972

The first system to pass a version of the Turing test

Psychiatrists couldn't distinguish interviews with PARRY from (text transcripts of) interviews with people diagnosed with paranoid schizophrenia

- Colby, K. M., Hilf, F. D., Weber, S., and Kraemer, H. C. (1972). Turing-like indistinguishability tests for the validation of a computer simulation of paranoid processes. *Artificial Intelligence* 3, 199–221.

ALICE and AIML

Based on ELIZA-like architecture, AIML

<https://pandorabots.com/docs/aiml-fundamentals/>

Richard Wallace put ALICE online, then reviewed interactions

If ALICE gave a poor response, Wallace added rule(s) to fix this

Over time, ALICE built a very large pattern-list of plausible replies

ALICE won the Loebner prize competition three times in 2000, 2001, and 2004.

Limited-domain QA chatbots, eg Hubert

For some practical applications, a limited QA chatbot is sufficient.
Eg FAQchat answers Frequently Asked Questions given an FAQ list.

B Abu Shawa and E Atwell. 2007. Chatbots: are they really useful?

Hubert: student feedback interviews, job applicant interviews

N Abbas et al. 2021. University student surveys using chatbots: AI conversational agents

Chatbots and Dialogue Systems

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Chatbots and Dialogue Systems

Corpus-based Chatbots

Two architectures for corpus-based chabots

Response by retrieval

- Use information retrieval to grab a response (that is appropriate to the context) from some corpus

Response by generation

- Use a language model or encoder-decoder to generate the response given the dialogue context

What conversations to draw on?

Spoken text in corpora: British National Corpus BNC, International Corpus of English ICE, Korpus Gesproke Afrikaans KGA

Transcripts of telephone conversations between volunteers

- Switchboard, CallHome corpora of telephone conversations

Question+Answer data-sets, FAQchat, IT helpdesk

Movie dialogue

- Various corpora of movie subtitles

Hire human crowdworkers to have conversations on a topic

Pseudo-conversations from public posts on social media

- Drawn from Twitter, Reddit, Weibo, etc.
- Tend to be noisy; often used just as pre-training.

Crucial to remove personally identifiable information (PII)

Response by retrieval: classic IR method

1. Given a user turn q , and a training corpus C of conversation
2. Find in C the turn r that is most similar (tf-idf cosine) to q
3. Say r
4. Or say the response to r (question+answer corpus)

Response by retrieval: neural IR method

1. Given a user turn q , and a training corpus C of conversation
2. Find in C the turn r that is most similar (BERT dot product) to q
3. Say r
4. Or say the response to r (question+answer corpus)

Response by generation

Think of response production as an encoder-decoder task

Generate each token r_t of the response by conditioning on the encoding of the entire query q and the response so far $r_1 \dots r_{t-1}$

$$\hat{r}_t = \operatorname{argmax}_{w \in V} P(w | q, r_1 \dots r_{t-1})$$

P.G. Wodehouse predicts neural chatbots

“What ho!” I said.

“What ho!” said Motty.

“What ho! What ho!”

“What ho! What ho! What ho!”

After that it seemed rather difficult to go on with the conversation.”

Wodehouse *My Man Jeeves*, 1919

Ongoing research problem: Neural chatbots can get repetitive and boring

A: Where are you going?

B: I'm going to the restroom.

A: See you later.

B: See you later.

A: See you later.

B: See you later.

A: How old are you?

B: I'm 16.

A: 16?

B: I don't know what you are talking about.

A: You don't know what you are saying.

B: I don't know what you are talking about.

A: You don't know what you are saying.

Response by retrieving and refining knowledge

Can generate responses from informative text rather than dialogue

- To respond to turns like “Tell me something about Beijing”
 - Xiaolce collects sentences from public lectures and news articles.
 - And searches them using IR based on query expansion from user’s turn

Chatbots: pro and con

Pro:

- Fun
- Good for narrow, scriptable applications

Cons:

- They don't really understand
- Giving the appearance of understanding may be problematic
- Rule-based chatbots are expensive and brittle
- IR-based chatbots can only mirror training data
- Next steps: integrating chatbot ability into frame-based agents

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Corpus-based Chatbots

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The Dialogue-State Architecture

Dialogue-State or Belief-State Architecture

A more sophisticated version of the frame-based architecture

- Has dialogue acts, more ML, better generation

The basis for modern research systems

Slowly making its way into industrial systems

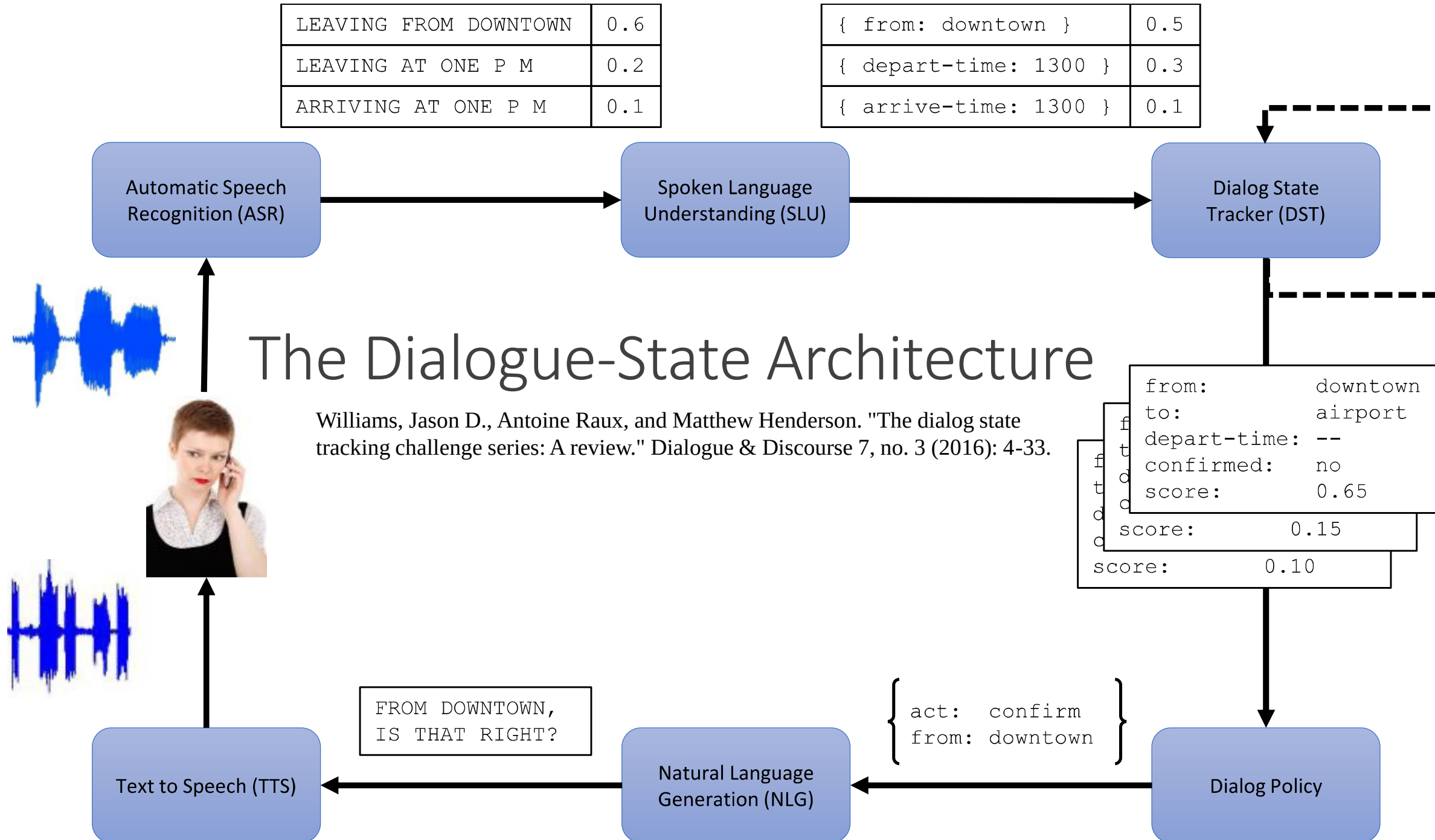
- Some aspects (ML for slot-understanding) already widely used industrially

LEAVING FROM DOWNTOWN	0.6
LEAVING AT ONE P M	0.2
ARRIVING AT ONE P M	0.1

{ from: downtown }	0.5
{ depart-time: 1300 }	0.3
{ arrive-time: 1300 }	0.1

The Dialogue-State Architecture

Williams, Jason D., Antoine Raux, and Matthew Henderson. "The dialog state tracking challenge series: A review." *Dialogue & Discourse* 7, no. 3 (2016): 4-33.



Components in a dialogue-state architecture

NLU: extracts slot fillers from the user's utterance using machine learning

Dialogue state tracker: maintains the current state of the dialogue (user's most recent dialogue act, set of slot-filler constraints from user)

Dialogue policy: decides what the system should do or say next

- GUS policy: ask questions until the frame was full then report back
- More sophisticated: know when to answer questions, when to ask a clarification question, etc.

NLG: produce more natural, less templated utterances

The task of dialogue state tracking

User: I'm looking for a cheaper restaurant

`inform(price=cheap)`

System: Sure. What kind - and where?

User: Thai food, somewhere downtown

`inform(price=cheap, food=Thai, area=centre)`

System: The House serves cheap Thai food

User: Where is it?

`inform(price=cheap, food=Thai, area=centre); request(address)`

System: The House is at 106 Regent Street

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Evaluating Dialogue Systems

Evaluating chatbots and task-based dialogue

Task-based dialogue:

- mainly by measuring task performance

Conversational chatbots:

- mainly by human evaluation

Chatbots are evaluated by humans

Participant evaluation: The human who talked to the chatbot assigns a score

Observer evaluation: third party who reads a transcript of a human/chatbot conversation assigns a score.

Participant evaluation

Human chats with model for 6 turns and rates 8 dimensions of quality:

- **avoiding repetition, interestingness, making sense, fluency, listening, inquisitiveness, humanness, engagingness,**

(1) Avoiding Repetition: How repetitive was this user?

- •Repeated themselves over and over •Sometimes said the same thing twice • Always said something new

(3) Making sense: How often did this user say something which didn't make sense?

- •Never made any sense •Most responses didn't make sense •Some responses didn't make sense •Everything made perfect sense

(8) Engagingness: How much did you enjoy talking to this user?

- •Not at all •A little •Somewhat •A lot

Observer evaluation: acute-eval

Li, M., Weston, J., and Roller, S. (2019). Acute-eval: Improved dialogue evaluation with optimized questions and multi-turn comparisons. *NeurIPS19 Workshop on Conversational AI*.

Annotators look at two conversations (A + B) and decide which is better:

Engagingness: Who would you prefer to talk to for a long conversation?

Interestingness: If you had to say one of these speakers is interesting and one is boring, who would you say is more interesting?

Humanness: Which speaker sounds more human?

Knowledgeable: If you had to say that one speaker is more knowledgeable and one is more ignorant, who is more knowledgeable?

Automatic evaluation is an open problem

Automatic evaluation methods (like the BLEU scores used for Machine Translation) are generally not used for chatbots.

- They correlate poorly with human judgements.

One current research direction: **Adversarial Evaluation**

- Inspired by the Turing Test
- train a ``Turing-like'' classifier to distinguish between human responses and machine responses.
- The more successful a dialogue system is at fooling the evaluator, the better the system.

More fine-grained metrics: User Satisfaction Survey

Walker, Marilyn, Candace Kamm, and Diane Litman. "Towards developing general models of usability with PARADISE." *Natural Language Engineering* 6, no. 3 & 4 (2000): 363-377.

TTS Performance	Was the system easy to understand ?
ASR Performance	Did the system understand what you said?
Task Ease	Was it easy to find the message/flight/train you wanted?
Interaction Pace	Was the pace of interaction with the system appropriate?
User Expertise	Did you know what you could say at each point?
System Response	How often was the system sluggish and slow to reply to you?
Expected Behavior	Did the system work the way you expected it to?
Future Use	Do you think you'd use the system in the future?

Other Heuristics

Efficiency cost:

- total elapsed time for the dialogue in seconds,
- the number of total turns or of system turns
- total number of queries
- “turn correction ratio”: % of turns that were used to correct errors

Quality cost:

- number of ASR rejection prompts.
- number of times the user had to barge in

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Chatbots and Ethical Issues

Dialog System Design: User-centered Design

1. Study the users and task
2. Build simulations
 - **Wizard of Oz** study
3. Iteratively test design on users



Ethical design

Ethical issues have long been known to be crucial in artificial agents

Mary Shelley's *Frankenstein*

- creating agents without a consideration of ethical and humanistic concerns

Ethical issues:

- **Safety:** Systems abusing users, distracting drivers, or giving bad medical advice
- **Representational harm:** Systems demeaning particular social groups
- **Privacy:** Information Leakage



Safety

Chatbots for mental health

- Extremely important not to say the wrong thing

In-vehicle conversational agents

- Must be aware of environment, driver's level of attention

Abuse and Representation Harm: The case of Microsoft Tay

Experimental Twitter chatbot launched in 2016

- given the profile personality of an 18- to 24-year-old American woman
- could share horoscopes, tell jokes,
- asked people to send selfies
- used informal language, slang, emojis, and GIFs,
- Designed to learn from users (IR-based)

The case of Microsoft Tay

Immediately Tay turned offensive and abusive

- Obscene and inflammatory tweets
- Nazi propaganda, conspiracy theories
- Began harassing women online
- Reflecting racism and misogyny of Twitter users

Microsoft took Tay down after 16 hours

Lessons:

- User response must be considered in the design phase

Bias in training datasets

Henderson *et al.* ran hate-speech and bias detectors on standard training sets for dialogue systems:

- Twitter, Reddit, other dialogue datasets

Found bias and hate-speech

- In training data
- In dialogue models trained on the data

Privacy: Training on user data

Accidental information leakage

- “Computer, turn on the lights [answers the phone] Hi, yes, my password is...”
- Henderson show in simulation that this leakage can occur.

Intentional information leakage

- Dialogue systems that are designed to send user data to developer or advertisers
- Important to consider privacy-preserving dialogue systems

Chatbots and
Dialogue
Systems

Chatbots and Ethical Issues

Chatbots and Dialogue Systems

Introduction to Chatbots and
Dialogue Systems

Properties of Human
Conversation

Rule-based Chatbots: ELIZA,
PARRY, AIML, ALICE, HUBERT

Corpus-based Chatbots

The Dialogue-State Architecture

Evaluating Dialogue Systems

Chatbots and Ethical Issues